

Beautifframe

Beautiful Theorem-Like Environments for Typst

Version 0.3.1

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1 Introduction

Beautiframing is a Typst package for creating beautiful theorem-like environments (theorems, definitions, lemmas, proofs, etc.) with multiple visual styles.

1.1 Features

- **9 distinct styles:** classic, modern, elegant, colorful, boxed, minimal, academic, **bw**, **cours**
- **6 variants per style:** prominent, standard, subtle, accent, minimal, inline
- **Flexible mapping:** Assign any variant to any environment type
- **Independent counters:** Each environment type has its own counter
- **Customizable labels:** Change “Theorem” to “Théorème”, “Satz”, etc.
- **QED symbol presets:** □, ■, ▀, CQFD,
- **Color themes:** Pre-built themes (ocean, forest, sunset, lavender)
- **Language presets:** French, German, Spanish
- **French Math Preset:** one-call setup for French secondary math courses
- **QR sidebar:** attach a QR code column to any environment
- **Environment references:** label theorem-like blocks and link back to their page
- **Student fill space:** blank, ruled lines, or dot grid appended inside any environment
- **Print-friendly modes:** color, grayscale, black & white

1.2 Quick Start

```
#import "@preview/beautiframing:0.3.1": *

#theorem(name: "Pythagorean")[
  In a right triangle:  $a^2 + b^2 = c^2$ 
]

#definition[
  A *limit* is the value that a function approaches.
]

#proof[
  The proof is left as an exercise.
]
```

1.2.1 French Math Quick Start

```
#import "@preview/beautiframing:0.3.1": *

#preset-french-math() // or #preset-french-math-bw()

#theoreme(name: "Pythagore")[Dans un triangle rectangle:  $a^2 + b^2 = c^2$ ]
#definitionfr[Une fonction continue préserve les limites.]
#pratique(space: "lines", space-height: 3cm)[Calculer la dérivée de  $f(x) = x^3$ .]
```

2 Environments

2.1 Available Environments

Beautiframed provides 8 environment types:

Environment	Default Variant	Counter	Usage
theorem	prominent	Optional	Main results
definition	standard	Optional	Foundational concepts
lemma	standard	Optional	Supporting results
proposition	standard	Optional	Secondary results
corollary	standard	Optional	Consequences
remark	subtle	Optional	Commentary
example	accent	Optional	Illustrations
proof	(special)	No	Demonstrations

2.2 Basic Usage

Each environment function accepts:

- **name:** Optional name (e.g., "Pythagorean" or [Pythagorean])
- **title:** Synonym for **name** — both are accepted, **name** takes priority
- **number:** auto (default), none, or custom value
- **label:** Typst label for env-ref, for example <thm-pythagore>
- **qr:** URL string to attach a QR code sidebar (requires qr-renderer configured)
- **space:** "empty", "lines", or "grid" — adds fill space for student work
- **space-height:** Height of the fill area (default 3cm)
- **body:** The content

```
#theorem(name: "Fermat's Last")[
  There are no positive integers  $a$ ,  $b$ ,  $c$  such that
   $a^n + b^n = c^n$  for  $n > 2$ .
]
```

```
// title: is accepted as a synonym for name:
#theorem(title: "Fermat's Last")[
  There are no positive integers  $a$ ,  $b$ ,  $c$  such that
   $a^n + b^n = c^n$  for  $n > 2$ .
]
```

Important: When the name contains math, use content syntax [...] instead of string "...", because strings do not process math markup:

```
// Wrong –  $f$  appears as literal text
#theorem(name: "Continuity of  $f$ ")[...]

// Correct
#theorem(name: [ $f$  is continuous]) [...]
```

Theorem 1 | There are no positive integers a , b , c such that $a^n + b^n = c^n$ for $n > 2$.
Fermat's Last

2.3 Numbering Control

2.3.1 Automatic Numbering (default)

```
#theorem[First theorem] // Theorem 1
#theorem[Second theorem] // Theorem 2
```

2.3.2 No Numbering

```
#theorem(number: none)[Unnumbered theorem]
#definition(number: none)[Unnumbered definition]
```

Theorem 1 | This theorem has no number.

Example

2.3.3 Custom Numbering

```
#theorem(number: "A")[Special theorem A]
#theorem(number: "★")[Starred theorem]
```

Theorem A | Special theorem A

2.3.4 Numbered Remarks

By default, remarks have no number. To number them:

```
#remark(number: auto)[This remark will be numbered.]
```

Remark 1 | This remark is numbered.

Remark | This remark is not numbered (default).

2.4 References

Add a Typst label to any environment, then reference it with `#env-ref(<label>)`. The reference text includes the environment label, number, and target page, and the whole text links to the labelled block.

```
#theorem(label: <thm-pythagore>, title: "Pythagore")[
  Dans un triangle rectangle:  $a^2 + b^2 = c^2$ .
]
```

```
Voir #env-ref(<thm-pythagore>).
// -> Théorème 1 (p. 3)
```

```
#remark(label: <rem-unites>)[Attention aux unités.]
Voir #env-ref(<rem-unites>).
// -> Remark (p. 3)
```

Use `page: false` to hide the page number:

```
#env-ref(<thm-pythagore>, page: false)
```

Use `env-refs` for several environments. Consecutive references with the same label are compacted:

```
#pratique(label: <prac-3>)[...]
#pratique(label: <prac-4>)[...]
#pratique(label: <prac-5>)[...]
#pratique(label: <prac-6>)[...]
```

```
Voir #env-refs(<prac-3>, <prac-4>, <prac-5>, <prac-6>, page: false).  
// -> En pratique 3-6
```

```
#definition(label: <def-limite>)[...]  
#proposition(label: <prop-limite>)[...]
```

```
Voir #env-refs(<def-limite>, <prop-limite>, page: false).  
// -> Définition 1 et Proposition 2
```

2.4.1 Counter Reset

Reset all counters manually:

```
#beautiframe-reset()
```

3 Styles

Beautiframe includes 9 visual styles. Each style provides 6 variants.

3.1 Classic Style

Traditional textbook layout with badge on the left and content with left border. Based on the exercise-bank pattern.

```
#beautiframe-setup(style: "classic")
#theorem(name: "Classic")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
#remark[This is the default style.]
```

Theorem 1 | In a right triangle: $a^2 + b^2 = c^2$
Classic

Definition 1 | A continuous function preserves limits.

Remark | This is the default style.

3.2 Modern Style

Contemporary design with thick accent bars and rule separators.

```
#beautiframe-setup(style: "modern")
#theorem(name: "Modern")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 — *Modern*

In a right triangle: $a^2 + b^2 = c^2$

Definition 1

A continuous function preserves limits.

3.3 Elegant Style

Sophisticated styling with centered headers and decorative ornaments.

```
#beautiframe-setup(style: "elegant")
#theorem(name: "Elegant")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

◇ **THEOREM 1** ◇

(Elegant)

In a right triangle: $a^2 + b^2 = c^2$

• **DEFINITION 1** •

A continuous function preserves limits.

3.4 Colorful Style

Each environment type has a distinct color for visual differentiation.

```
#beautiframed-setup(style: "colorful")
#theorem[Red for theorems.]
#definition[Blue for definitions.]
#example[Green for examples.]
```

Theorem 1

Red for theorems.

Definition 1

Blue for definitions.

Example 1

Green for examples.

3.5 Boxed Style

Full framed boxes with optional backgrounds and header bars.

```
#beautiframed-setup(style: "boxed", border-radius: 3pt)
#theorem(name: "Boxed")[In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition[A continuous function preserves limits.]
```

Theorem 1 (*Boxed*)

In a right triangle: $a^2 + b^2 = c^2$

Definition 1

A continuous function preserves limits.

3.6 Minimal Style

Ultra-clean, print-friendly. Minimal ink usage.

```
#beautiframed-setup(style: "minimal")
#theorem(name: "Minimal") [In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition [A continuous function preserves limits.]
```

Theorem 1 (*Minimal*).

In a right triangle: $a^2 + b^2 = c^2$

Definition 1. A continuous function preserves limits.

3.7 Academic Style

Formal research paper style matching AMS/journal conventions.

```
#beautiframed-setup(style: "academic")
#theorem(name: "Academic") [In a right triangle:  $a^2 + b^2 = c^2$ ]
#definition [A continuous function preserves limits.]
```

THEOREM 1 (*Academic*).

In a right triangle: $a^2 + b^2 = c^2$

Definition 1.

A continuous function preserves limits.

3.8 BW Style

Black-and-white two-column layout for printed French math courses (Gymnomath / cours Collège). A 3.35 cm right-aligned label column is separated from the content column by a horizontal rule (standard variant) or a box border (boxed/prominent variants).

Variants: **standard** (side-block with rule), **boxed** (light rect), **prominent** (thicker rect, for théorèmes), **accent** (env-color stroke and label), **minimal** (inline label), **inline**, **proof** (with QED).

```
#preset-french-math-bw()
#theoreme(name: "Pythagore") [Dans un triangle rectangle:  $a^2 + b^2 = c^2$ ]
#definitionfr [Une fonction continue préserve les limites.]
#remarque [La réciproque est généralement fausse.]
#preuve [Par application directe de la définition.]
```

Théorème 1
Pythagore

Dans un triangle rectangle: $a^2 + b^2 = c^2$

Définition 1 Une fonction continue préserve les limites.

Remarque La réciproque est généralement fausse.

Preuve. Par application directe de la définition. □

3.9 Cours Style

French course style with a 2 cm label column that overhangs 1 cm into the left margin, and content framed with a thin left border. Optimised for A4 course sheets with 2.2 cm margins. Blue accent color by default.

Variants: **standard** (accent-colored border and label), **accent** (per-env color), **subtle** (muted), **minimal** (inline, for remarques), **inline**, **proof** (with QED).

```
#preset-french-math()
#theoreme(name: "Valeurs intermédiaires")[
  Si  $f$  est continue sur  $[a ; b]$ , pour tout  $y$  entre  $f(a)$  et  $f(b)$  il existe
   $c$  tel que  $f(c) = y$ .
]
#definitionfr[Une suite converge si elle admet une limite finie.]
#remarque[La réciproque est fausse en général.]
```

Théorème 1 | Si f est continue sur $[a; b]$, pour tout y entre $f(a)$ et $f(b)$ il existe c tel que $f(c) = y$.
Valeurs intermédiaires

Définition 1 | Une suite converge si elle admet une limite finie.

Remarque La réciproque est fausse en général.

4 Variants

Each style provides variants that can be assigned to any environment type. The 6 core variants are available in all styles:

Variant	Description	Default for
prominent	Strongest visual emphasis, thick borders, bold colors	theorem
standard	Normal, balanced styling	definition, lemma, proposition, corollary
subtle	Lighter, less prominent, muted colors	remark
accent	Uses environment-specific color	example
minimal	Very light styling, minimal visual elements	—
inline	No structural elements, flows with text	—

Boxed style has 3 additional variants:

Variant	Description
titled	Label breaks the top border line (beautitled-style)
centered	Label centered at top, breaking the border
corner	L border (top + left sides)
corner2	Inverted L border (left + bottom sides)

4.1 Assigning Variants

Map any variant to any environment type:

```
#beautifframe-setup(
  theorem-variant: "prominent", // Theorems get maximum emphasis
  lemma-variant: "subtle",      // Lemmas are de-emphasized
  remark-variant: "inline",     // Remarks flow inline
  example-variant: "accent",    // Examples use their color
)
```

4.2 Setting All Variants at Once

The `default-variant` parameter sets **all** 7 environment variants in a single call. Individual overrides are applied after, so they win:

```
// Make every environment use the boxed variant
#beautifframe-setup(default-variant: "boxed")

// All boxed, except theorems which get prominent
#beautifframe-setup(default-variant: "boxed", theorem-variant: "prominent")
```

This is useful when switching styles (e.g., `bw` → `cours`) and wanting a uniform appearance, or when preparing a worksheet where all environments should have the same fill-space variant.

All minimal via default-variant:

Théorème 1 Content.
Sample

Définition 1 Content.

Remarque Content.

All **standard**, theorem overridden to **prominent**:

Théorème 1 | Content.

Sample

Définition 1 | Content.

Remarque | Content.

4.3 Visual Gallery: All Variants × All Styles

4.3.1 Classic Style

Prominent:

Théorème 1 | Sample text.
Name

Subtle:

Théorème 3 | Sample text.
Name

Minimal:

Théorème 5 Sample text.
Name

Standard:

Théorème 2 | Sample text.
Name

Accent:

Théorème 4 | Sample text.
Name

Inline:

Théorème 6 (*Name*). Sample text.

4.3.2 Modern Style

Prominent:

Théorème 1 — *Name*

Sample text.

Subtle:

Théorème 3 — *Name*

Sample text.

Minimal:

Théorème 5 — *Name*

Sample text.

Standard:

Théorème 2 — *Name*

Sample text.

Accent:

Théorème 4 — *Name*

Sample text.

Inline:

Théorème 6 — *Name* — Sample text.

4.3.3 Elegant Style

Prominent:

◇

THÉORÈME 1

◇

(Name)

Sample text.

Standard:

•

THÉORÈME 2

•

(Name)

Sample text.

Subtle:

Théorème 3

(Name).

Sample text.

Accent:

★

THÉORÈME 4

★

(Name)

Sample text.

Minimal:

THÉORÈME 5

(Name)

Sample text.

Inline:

THÉORÈME 6

(Name).

Sample text.

4.3.4 Colorful Style

Prominent:

Théorème 1 (*Name*)

Sample text.

Subtle:

Théorème 3 (*Name*)

Sample text.

Minimal:

Théorème 5 (*Name*)

Sample text.

Standard:

Théorème 2 (*Name*)

Sample text.

Accent:

Théorème 4 (*Name*)

Sample text.

Inline:

Théorème 6 (*Name*). Sample text.

4.3.5 Boxed Style

Prominent:

Théorème 1 (*Name*)

Sample text.

Standard:

Théorème 2 (*Name*)

Sample text.

Subtle:

Théorème 3 (*Name*)

Sample text.

Accent:

Théorème 4 (*Name*)

Sample text.

Minimal:

Théorème 5 (*Name*)

Sample text.

Inline:

[Théorème 6: *Name*] Sample text.

Titled:

Théorème 7 (*Name*)

Sample text.

Centered:

Théorème 8 (*Name*)

Sample text.

Corner:

Théorème 9 (*Name*)

Sample text.

Corner2:

Théorème 10 (*Name*)

Sample text.

4.3.6 Minimal Style

Prominent:

Théorème 1 (*Name*).

Sample text.

Subtle:

Théorème 3 (*Name*). Sample text.

Minimal:

Théorème 5. Sample text.

Standard:

Théorème 2 (*Name*). Sample text.

Accent:

THÉORÈME 4 (*Name*). Sample text.

Inline:

(**Théorème 6:** *Name*) Sample text.

4.3.7 Academic Style

Prominent:

THÉORÈME 1 (*Name*).

Sample text.

Subtle:

Théorème 3 (*Name*). Sample text.

Standard:

Théorème 2 (*Name*).

Sample text.

Accent:

THÉORÈME 4 (*Name*).

Sample text.

Minimal:

Théorème 5. Sample text.

Inline:

Théorème 6 (*Name*): Sample text.

4.3.8 BW Style

Prominent:

Théorème 1

Name

Sample text.

Standard:

Théorème 2

Name

Sample text.

Boxed:

Théorème 3

Name

Sample text.

Accent:

Théorème 4

Name

Sample text.

Minimal:

Théorème 5

(Name)

Sample text.

Inline:

Théorème 6

(Name).

Sample text.

4.3.9 Cours Style

Standard:

Théorème 1 | Sample text.
Name

Subtle:

Théorème 3 | Sample text.
Name

Inline:

Théorème 5 (*Name*). Sample text.

Accent:

Théorème 2 | Sample text.
Name

Minimal:

Théorème 4 Sample text.
Name

5 Proofs and QED Symbols

5.1 Basic Proof

```
#proof[
```

```
    By direct calculation, we have  $2^2 = 4$ .
```

```
]
```

Preuve. By direct calculation, we have $2^2 = 4$.

□

5.2 QED Symbol Presets

Beautiframe provides several QED symbol presets:

```
#qed-square()      // □ (default)
#qed-filled()      // ■
#qed-tombstone()   // ■
#qed-cqfd()        // CQFD (French)
#qed-slashes()     // //
#qed-text()        // Q.E.D.
#qed-none()        // (no symbol)
```

Default (□):

Preuve. Hollow square.

□

Filled (■):

Preuve. Filled square.

■

Tombstone (■):

Preuve. Tombstone symbol.

■

CQFD:

Preuve. French style.

CQFD

Slashes:

Q.E.D.:

Preuve. Double slash.

//

Preuve. Latin abbreviation.

Q.E.D.

5.3 Custom QED Symbol

```
#beautiframe-setup(qed-symbol: text(fill: green, sym.checkmark))
```

Preuve. Custom green checkmark.

✓

6 Colors and Themes

6.1 Global Colors

```
#beautiframe-setup(
  primary-color: rgb("#1a5276"),
  secondary-color: rgb("#7f8c8d"),
  accent-color: rgb("#2980b9"),
)
```

6.2 Per-Environment Colors

Used primarily by the colorful style:

```
#beautiframe-setup(
  theorem-color: rgb("#c0392b"), // Red
  definition-color: rgb("#2980b9"), // Blue
  lemma-color: rgb("#8e44ad"), // Purple
  example-color: rgb("#27ae60"), // Green
  remark-color: rgb("#7f8c8d"), // Gray
)
```

6.3 Color Themes

Pre-built color schemes:

Ocean Theme:

Théorème

Blue tones throughout.

Forest Theme:

Théorème

Green tones throughout.

Sunset Theme:

Théorème

Warm red and orange tones.

Lavender Theme:

Théorème

Purple tones throughout.

6.4 Print-Friendly Modes

For B&W printing:


```
#beautiframe-setup(color-mode: "color")    // Full color (default)
#beautiframe-setup(color-mode: "grayscale") // Grayscale
#beautiframe-setup(color-mode: "bw")       // Pure black and white
```

Color mode:**Théorème**

Full color styling.

Grayscale mode:**Théorème**

Grayscale for B&W printers.

B&W mode:**Théorème**

Pure black and white.

7 Language Presets

7.1 French

`#preset-french()`

Théorème 1 | Dans un triangle rectangle: $a^2 + b^2 = c^2$
Pythagore

Définition 1 | Une fonction continue préserve les limites.

Preuve. Immédiat. □

7.2 German

`#preset-german()`

Satz 1 | In einem rechtwinkligen Dreieck: $a^2 + b^2 = c^2$
Pythagoras

Definition 1 | Eine stetige Funktion erhält Grenzwerte.

Beweis. Offensichtlich. □

7.3 Spanish

`#preset-spanish()`

Teorema 1 | En un triángulo rectángulo: $a^2 + b^2 = c^2$
Pitágoras

Definición 1 | Una función continua preserva límites.

Demostración. Inmediato. □

7.4 Custom Labels

```
#beautiframe-setup(
  theorem-label: "Théorème",
  definition-label: "Définition",
  proof-label: "Démonstration",
)
```

8 Layout Configuration

8.1 Classic Style Layout

The classic style uses a grid layout with configurable dimensions:

```
#beautiframing-setup(
  line-position: 2cm,    // Distance from left to vertical line
  label-extra: 1cm,     // Extension into left margin
  border-width: 1.5pt,  // Line thickness
)
```

8.2 Spacing

```
#beautiframing-setup(
  theorem-above: 1.2em, // Space before theorems
  theorem-below: 1em,   // Space after theorems
  header-gap: 0.4em,    // Gap between label and body
)
```

8.3 Typography

Control label appearance:

```
#beautiframing-setup(
  label-weight: "bold", // "bold", "regular", "semibold", etc.
  label-size: 11pt,    // Size of "Theorem", "Definition", etc.
  name-style: "italic", // Style for theorem names: "italic" or "normal"
)
```

Bold label (default):

Teorema 1 (*Name*)

Sample text.

Regular label:

Teorema 2 (*Name*)

Sample text.

8.4 Boxed Style Options

```
#beautiframing-setup(
  inset: (x: 1em, y: 0.8em), // Padding inside boxes
  border-radius: 4pt,         // Rounded corners
  border-width: 1.5pt,        // Border thickness
)
```

9 Custom Environments

Create your own environment types with independent counters using `new-env`:

```
// Create custom environments
#let conjecture = new-env("Conjecture", base: "theorem")
#let propriete = new-env("Propriété", base: "definition", numbered: false)
#let formule = new-env("Formule", base: "lemma", color: green)
#let axiom = new-env("Axiom", base: "theorem", numbered: true)
```

```
// Use them like built-in environments
#conjecture[Every even number greater than 2 is the sum of two primes.]
#conjecture(name: "Goldbach")[Famous unsolved problem.]
#propriete[A property without number.]
#formule[The quadratic formula:  $x = \frac{-b \pm \sqrt{b^2 - 4a c}}{2a}$ ]
```

Conjecture 1

Every even number greater than 2 is the sum of two primes.

Conjecture 2 (*Goldbach*)

Famous unsolved problem in number theory.

Axiom 1

Two points determine a unique line.

9.1 Parameters

```
#let my-env = new-env(
  "Label",          // Display label (required)
  base: "theorem",  // Inherit style from: theorem, definition, lemma, etc.
  numbered: true,    // Auto-number by default
  color: none,       // Optional custom color
)
```

9.2 Plural Forms

All environments support a `plural` parameter. Default plurals are provided for English and all language presets:

```
#theorem(plural: true)[Multiple theorems grouped together.]
#definition(plural: true)[Several related definitions.]
```

Teorema 1

A single theorem.

Teoremas 2

Multiple theorems can be grouped in one box.

Language presets automatically set the correct plural forms:

```
#preset-french() // Sets: Théorèmes, Définitions, Lemmes, etc.
#preset-german() // Sets: Sätze, Definitionen, Lemmata, etc.
#preset-spanish() // Sets: Teoremas, Definiciones, Lemas, etc.
```

Custom environments also support plurals:

```
#let conjecture = new-env("Conjecture", plural: "Conjectures", base: "theorem")
#conjecture(plural: true)[Two famous conjectures.]
```

9.3 Resetting Custom Counters

```
#reset-env("Conjecture") // Reset specific custom environment
#beautiframe-reset()     // Reset all built-in counters
```

10 French Math Preset

The French math preset configures beautiframe in a single call for French secondary and post-secondary math courses.

10.1 Preset Functions

```
// Color version – cours style, blue accent, bold labels, QED square
#preset-french-math()
```

```
// Black-and-white version – bw style, 8.4pt labels, luma palette
#preset-french-math-bw()
```

```
// Reset all counters including custom French environments
#beautiframe-reset-french-math()
```

10.2 French Environments

All French environments accept `name:` / `title:` (alias), `qr:`, `space:`, `space-height:`.

Function	Label	Base	Numbered
theoreme	Théorème	theorem	Yes
definitionfr	Définition	definition	Yes
propositionfr	Proposition	proposition	Yes
exemplefr / exemple	Exemple	example	Yes
remarque	Remarque	remark	No
corollaire	Corollaire	corollary	Yes
preuve	Preuve	proof	No
pratique	En pratique	example	Yes
propriete	Propriété	corollary	No
formule	Formule	lemma	Yes
formules(...)	Formules	lemma	Yes
methode	Méthode	proposition	Yes
notation(...)	Notation	remark	No
discussion(...)	Discussion	remark	No
objectifs / objectif	Objectifs d'apprentissage	lemma	No
concepts	Concepts clés	lemma	No
glossaire	Glossaire	lemma	No
defi / défi	 Défi	remark	No

10.3 Challenge Callout (defi)

`#defi` renders a compact, unnumbered challenge callout using the remark style. The icon and title are optional.

| Parameter | Type | Default | Description | |———|———|———|———| | **title** | content or none | none | Optional title appended after “Défi :” | | **icon** | content or none | [] | Decorative icon; set none to hide | | **label** | label or none | none | Cross-reference label | | **qr** | string or none | none | QR sidebar URL | | **space** | string or none | none | Fill space type | | **space-height** | length | 3cm | Height of fill area | | **body** | content | — | Challenge content |

```
#defi[Résoudre  $x^2 - 5x + 6 = 0$ .]
```

```
#defi(title: [Y arrivez-vous ?])[Montrer que  $\sqrt{2}$  est irrationnel.]
```

```
#defi(icon: [🍌])[Justifier chaque étape du raisonnement.]
#defi(icon: none)[Sans icône.]
```

🎯 Défi Résoudre $x^2 - 5x + 6 = 0$.

🎯 Défi : Y arrivez-vous ? Montrer que $\sqrt{2}$ est irrationnel.

10.4 Formula Recap (`formule-end` / `formules-recap`)

Collect formulas throughout a chapter and print them all at once in a recap box at the end. Use `#formule-end` to register a formula (nothing is displayed at the call site), then `#formules-recap` to render the collected list.

```
// Register formulas anywhere in the chapter (nothing rendered in place)
#formule-end([Discriminant], $Delta = b^2 - 4a c$)
#formule-end([Racines], $x = (-b plus.minus sqrt(Delta)) / (2a)$)
```

```
// Render the recap (clears the list by default)
#formules-recap()
```

```
// Custom title
#formules-recap(title: [Formules du chapitre])
```

```
// Keep the list for a second recap (clear: false)
#formules-recap(clear: false)
```

Parameter	Type	Default	Description	label	content
<code>formula</code>	string	bold	Formula name displayed in bold	<code>formula</code>	content
<code>title</code>	string	[Formules à retenir]	Box header	<code>clear</code>	bool true
<code>clear</code>	bool	true	Clear the collection after printing		

Formules à retenir | Discriminant :

$$\Delta = b^2 - 4ac$$

Racines :

$$x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}$$

10.5 Course Meta-Environments

Three unnumbered environments intended for course structure:

- `#objectifs` / `#objectif` — learning objectives at the start of a chapter
- `#concepts` — key concepts summary
- `#glossaire` — glossary or vocabulary list

```
#objectifs[
  - Savoir résoudre une équation du second degré
  - Connaître la formule du discriminant
]
```

```
#concepts[Équation, discriminant, racine, trinôme.]
```

```
#glossaire[*Racine* : valeur de $x$ pour laquelle $f(x) = 0$.]
```

- Objectifs d'apprentissage** |
- Savoir résoudre une équation du second degré.
 - Connaître et appliquer la formule du discriminant.

Concepts clés | Équation, discriminant, racine, trinôme du second degré.

10.6 Example

```
#import "@preview/beautiframe:0.3.1": *
#preset-french-math()

#theoreme(name: "Pythagore")[
  Dans tout triangle rectangle d'hypoténuse $c$: $a^2 + b^2 = c^2$.
]

#definitionfr[Une suite converge si elle admet une limite finie.]

#propriete[Toute suite monotone et bornée converge.]

#formule[
  $x = \displaystyle((-b \text{ plus.minus sqrt}(b^2 - 4a \ c)) / (2a))$
]

#pratique(space: "lines", space-height: 3cm)[
  Résoudre $2x^2 - 5x + 3 = 0$.
]

#preuve[Par calcul direct avec le discriminant.]
```

Théorème 1 | Dans tout triangle rectangle d'hypoténuse c : $a^2 + b^2 = c^2$.
Pythagore

Définition 1 | Une suite converge si elle admet une limite finie.

Propriété | Toute suite monotone et bornée converge.

Formule 1 |
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Notation On note f' la dérivée de f , et $f^{(n)}$ la dérivée n -ième.

Discussion La condition $a > 0$ est nécessaire mais pas suffisante.

En pratique 1 | Résoudre $2x^2 - 5x + 3 = 0$.

Preuve. Par calcul direct avec le discriminant.

□

11 QR Sidebar

Any environment can display a rendered QR code (or any content) in a right sidebar column.

11.1 Configuration

```
// Configure once in your preamble
#beautiframing-setup(
  qr-renderer: url => image.decode(
    tiaoma.qrcode(url, options: (border: 0)),
    format: "svg", width: 1.85cm
  ),
  qr-width: 1.85cm,
)
```

The `qr-renderer` key accepts a function `url => content`. When set to `none` (default), no sidebar is shown even if `qr:` is passed.

11.2 Usage

```
#theorem(qr: "https://example.com/proof")[
  In a right triangle:  $a^2 + b^2 = c^2$ .
]

#definitionfr(qr: "https://wiki.example.com/continuity")[
  Une fonction continue préserve les limites.
]
```

The QR sidebar works with all environments including custom ones created with `new-env`.

12 Student Fill Space

Append a blank fill area inside any environment for students to write their answers.

12.1 Parameters

- `space`: — "empty" (blank block), "lines" (8 mm ruled lines), "grid" (5 mm dot grid)
- `space-height`: — height of the fill area (default: 3cm)

12.2 Examples

// Blank area

```
#pratique(space: "empty", space-height: 2cm)[Define continuity.]
```

// Ruled lines

```
#pratique(space: "lines", space-height: 4cm)[Solve $2x - 5 = 7$.]
```

// Dot grid

```
#exemple(space: "grid", space-height: 5cm)[Sketch $f(x) = x^2$.]
```

Ruled lines (`space: "lines"`, height 3 cm):

En pratique 1 Calculer la dérivée de $f(x) = 3x^2 - 2x + 1$.

Dot grid (`space: "grid"`, height 3.5 cm):

Exemple 1 Représenter $f(x) = -x^2 + 4$ sur $[-3; 3]$.

```

. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . . .

```

13 API Reference

13.1 Environment Functions

```
// Built-in environments (all accept name:/title:, number:, qr:, space:, space-
height:)
#theorem(name: none, title: none, number: auto, qr: none, space: none, space-height:
3cm)[body]
#definition(name: none, title: none, number: auto, qr: none, space: none, space-
height: 3cm)[body]
#lemma(name: none, title: none, number: auto, qr: none, space: none, space-height:
3cm)[body]
#proposition(name: none, title: none, number: auto, qr: none, space: none, space-
height: 3cm)[body]
#corollary(name: none, title: none, number: auto, qr: none, space: none, space-
height: 3cm)[body]
#remark(name: none, title: none, number: none, qr: none, space: none, space-height:
3cm)[body]
#example(name: none, title: none, number: auto, qr: none, space: none, space-height:
3cm)[body]
#proof[body]

// French aliases
#theoreme      = theorem
#definitionfr  = definition
#propositionfr = proposition
#exemplefr     = exemple = example
#remarque      = remark
#corollaire    = corollary
#preuve(body)  = proof

// French custom environments
#pratique(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#propriete(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#formule(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#formules(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#methode(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#notation(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
#discussion(name: none, title: none, qr: none, space: none, space-height: 3cm)[body]
```

13.2 Setup Function

```
#beautiframe-setup(
  style: "classic",           // classic, modern, elegant, colorful, boxed,
                             // minimal, academic, bw, cours

  // — Variant mapping —
  // default-variant sets all 7 at once; individual params override it
  default-variant: none,     // e.g. "standard" – applies to all 7 types
  theorem-variant: "prominent",
  definition-variant: "standard",
  lemma-variant: "standard",
  proposition-variant: "standard",
  corollary-variant: "standard",
  remark-variant: "subtle",
  example-variant: "accent",

  // — Colors —
  primary-color: rgb("#2c3e50"),
  secondary-color: rgb("#7f8c8d"),
  accent-color: rgb("#2980b9"),
  background-color: white,
  // Per-environment colors (used by colorful style and accent variants)
  theorem-color: rgb("#c0392b"),
```

```

definition-color: rgb("#2980b9"),
lemma-color: rgb("#8e44ad"),
proposition-color: rgb("#8e44ad"),
corollary-color: rgb("#d35400"),
remark-color: rgb("#7f8c8d"),
example-color: rgb("#27ae60"),

// — Typography —————
label-size: 1em,           // 1em = body font size (recommended default)
label-weight: "bold",      // "bold", "regular", "semibold", etc.
name-style: "italic",      // subtitle style: "italic" or "normal"
body-size: none,          // inherited from document when none

// — Vertical spacing —————
theorem-above: 1em,
theorem-below: 0.8em,
definition-above: 1em,
definition-below: 0.8em,
lemma-above: 0.8em,
lemma-below: 0.6em,
proposition-above: 0.8em,
proposition-below: 0.6em,
corollary-above: 0.8em,
corollary-below: 0.6em,
remark-above: 0.6em,
remark-below: 0.6em,
example-above: 0.8em,
example-below: 0.8em,
proof-above: 0.5em,
proof-below: 0.8em,
header-gap: 0.3em,        // gap between label and body

// — Layout —————
inset: (x: 0.8em, y: 0.6em), // padding inside boxes
border-width: 1pt,           // stroke thickness
border-radius: 0pt,          // rounded corners (boxed style)
line-position: 2cm,          // vertical line distance from left (classic)
label-extra: 1cm,            // label overhang into left margin (classic/cours)

// — Numbering —————
numbering-format: "1",       // "1" or "1.1" (section.number)
link-to-section: false,      // prefix number with heading number
counter-reset: "manual",     // "manual", "section", "chapter"

// — Labels (singular) —————
theorem-label: "Theorem",
definition-label: "Definition",
lemma-label: "Lemma",
proposition-label: "Proposition",
corollary-label: "Corollary",
remark-label: "Remark",
example-label: "Example",
proof-label: "Proof",

// — Labels (plural, used when plural: true) —————
theorem-plural: "Theorems",
definition-plural: "Definitions",
lemma-plural: "Lemmas",
proposition-plural: "Propositions",
corollary-plural: "Corollaries",
remark-plural: "Remarks",
example-plural: "Examples",

// — QED —————
qed-symbol: sym.square.stroked,

// — Advanced —————
breakable: true,             // allow page breaks within environments
color-mode: "color",         // "color", "grayscale", "bw"

// — QR sidebar —————
qr-renderer: none,           // url => content function, or none to disable

```

```
    qr-width: 1.85cm,           // width of the QR sidebar column
)
```

13.3 Utility Functions

```
#beautiframe-reset()           // Reset all built-in counters
#beautiframe-reset-french-math() // Reset built-in + French env counters
#reset-env("Label")           // Reset a specific custom env counter

// French math presets
#preset-french-math()          // cours style, color, blue accent
#preset-french-math-bw()       // bw style, black-and-white

// Language presets
#preset-french()
#preset-german()
#preset-spanish()

// Color themes
#theme-ocean()
#theme-forest()
#theme-sunset()
#theme-lavender()

// QED presets
#qed-square()                 // □
#qed-filled()                 // ■
#qed-tombstone()              // ▀
#qed-cqfd()                   // CQFD
#qed-slashes()                // //
#qed-text()                   // Q.E.D.
#qed-none()                   // (none)
```